

Homework 1

First contact with Postgres and Pagila

Assigned	Tuesday, May 12, 2026
Due	Tuesday, May 19, 2026 - 6:00 PM (start of Week 2 class)
Submit on	Canvas — two files
Points	100 points (Part A: 50 · Part B: 50)
Time budget	~3 hours, including time spent verifying queries return reasonable results

What to Submit

Two files, both uploaded to Canvas:

1. **CS5200_HW1_lastname.pdf** - your written answers to Part A.
2. **CS5200_HW1_lastname.sql** - your SQL queries for Part B. Start from the provided CS5200_HW1_starter.sql file. I should be able to run the file top-to-bottom against a fresh Pagila database without edits. Test it before you submit.

Ground Rules

- **Talk it through, write it yourself.** Discussing problems with classmates is encouraged. Copying their answers or queries is not. The English-vs-SQL test from class applies: you should be talking English to each other, not SQL.
- **Don't rely on AI tools.** If an LLM is used through the acceptable use policy, make sure to disclose it at the top of the submitted file. See the AI policy in the Syllabus/Week 1 slides for what's in and out of bounds.
- **Stuck for an hour? Stop.** Email me, ask a classmate, set up an office hours virtual appointment. The worst use of your time on this assignment is silently grinding at midnight.
- **Don't forget the header:** Both files are required to have the standard header discussed during the lecture.

Part A: Concepts

50 points

Answer in your own words. Two to four sentences each for single-part questions, while multi-part questions should have a few sentences per part. I am looking for evidence the Week 1 lecture actually landed.

1. Database vs. DBMS (8 pts)

Briefly explain the difference between a database and a DBMS. Then give one example of each from your own life or work (anything you've used or heard of).

2. Data Independence (8 pts)

What is meant by data independence, and why was it considered a breakthrough when Codd proposed the relational model in 1970? (*Try to explain it the way you'd explain it to a non-technical friend.*)

3. AI policy in practice (8 pts)

Suppose, while working on Part B of this assignment, you got stuck for a long time on a question. You ask ChatGPT/Claude to explain why your query is returning the wrong results. Claude rewrites the query and gives you a working version that you don't fully understand.

- (a) According to the course AI policy, what specifically do you need to do before submitting the file?
- (b) What further step (or steps) would you take so that this is genuinely your work and not just the LLM's? Describe what you'd actually do and not what the policy says.

4. Read a database in the wild (12 pts)

Pick one application or service you use that obviously runs on a database. Not Pagila, and not the soccer-league example I'll introduce next week. Pick something real like Spotify, Venmo, your bank, MyChart, Instagram, or anything.

- (a) Name the application.
- (b) Name three pieces of information that you can reasonably guess are stored as columns of a single table somewhere in this system. (You don't need to know the schema, simply guess from the user-facing behavior.)
- (c) Name two relationships between tables you'd expect to exist. "Each X belongs to one Y" or "each Y can have many X" is fine.
- (d) Name one constraint or business rule the system probably enforces. (Examples: "a username can't be reused," "a balance can't go below zero unless overdraft is on," etc...)

5. Reading SQL (6 pts)

Below is a SELECT statement. In two or three sentences, describe what it returns in plain English and tell me what DESC changes about the output.

```
SELECT first_name, last_name, major, credits
FROM students
WHERE major ILIKE('Science')
      AND credits >= 60
ORDER BY last_name DESC
LIMIT 5;
```

6. Looking ahead (8 pts)

Pick one table from Pagila like film, actor, customer or any other table. Describe in plain English what each row in that table represents. Then name one other Pagila table that this one is related to, and describe the relationship. ("Each X is associated with one Y" / "Each X can have many Ys" / etc...) You don't need to know the schema. Explore the columns in DBeaver and determine the answer from what you see.

Part B: Ten Queries Against Pagila

50 points

Open the provided starter file CS5200_HW1_starter.sql in DBeaver (or your preferred client). Each of the 9 questions appears as a comment. Write a single SQL query under each one to answer it.

Constraints. Use only what we covered in Week 1: SELECT, FROM, WHERE, ORDER BY, LIMIT, DISTINCT, AND, basic comparison operators. The point of this assignment is reps with the basics.

Output. Each query should return only the columns the question asks for. For example, if the question asks for a film's title, don't also return the film_id.

Verifying your answers. Run each query. Eyeball the result. Does the row count look reasonable? Did the sort actually sort? You don't need to submit screenshots, but a query that returns ten thousand rows when the question expects three is a sign you read the question wrong.

Hints

- **Pagila stores most string values in uppercase.** Film titles like ACADEMY DINOSAUR, actor names like NICK WAHLBERG. If a query returns nothing and you used lowercase, that's why.
- **Postgres uses single quotes for string literals.** WHERE rating = 'PG', not WHERE rating = "PG". Double quotes mean something different in Postgres (they reference a column or identifier).
- **Sanity-check with LIMIT 5 first.** If a query feels uncertain, throw a LIMIT 5 on the end while you're iterating. Remove it once the query is correct.

How This Will Be Graded

Part A. Each question is graded on whether your answer demonstrates understanding of the concept. Two-sentence answers can earn full credit. Five-sentence answers that miss the point will not.

Part B. Each query is graded on a mix of different metrics. For example, if a query is worth 5 points, 4 will be graded for correctness (does it return the right results?) and 1 for cleanliness (right columns, right ordering, sensible formatting). I will run your file. Queries that error out get zero, queries that return wrong-but-close results can get partial.

An 'A' on this assignment is thoughtful, in-your-own-voice concept answers, and queries that work first-try. A 'C' could be partially correct queries or answers that just paraphrase the slides without engaging with them.